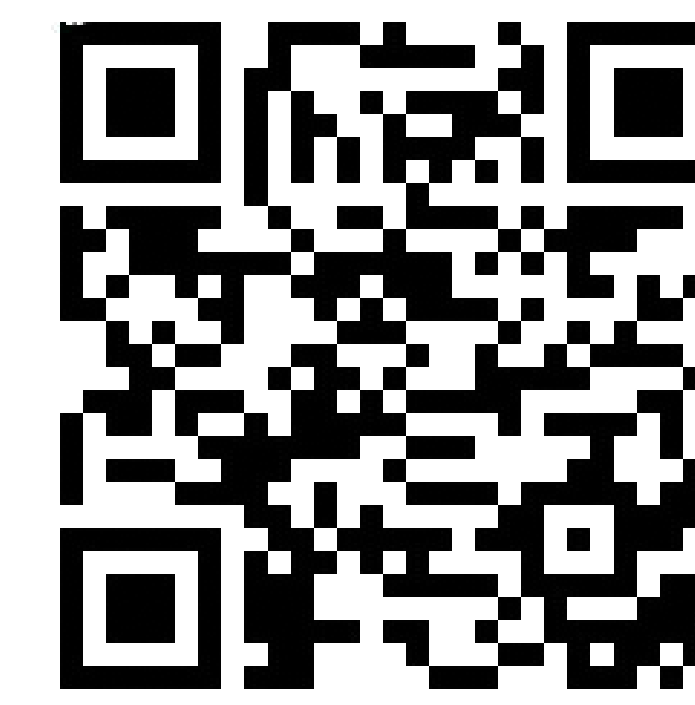




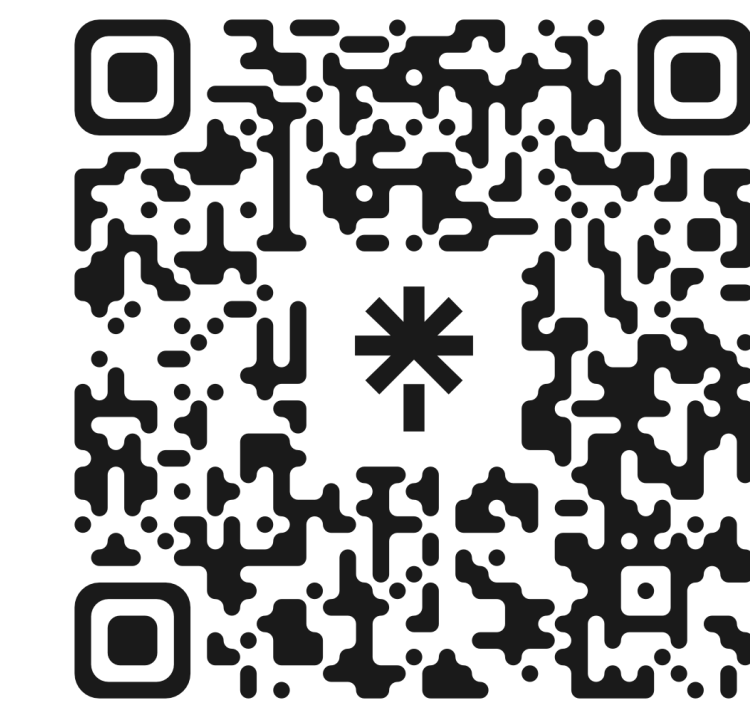
FIPRED: A study of predictive automated displays for wildfire fighting management.

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Full Paper



Linktree



Introduction

Mega wildfires such as the Los Angeles fires in the spring, the Hawaii Lahaina Fire in 2023, and the 2016 Fort McMurray Fire in Canada impose a devastating toll, destroying cities and taking lives.



Fort McMurray Fire (2016)

Fires often follow unexpected paths.

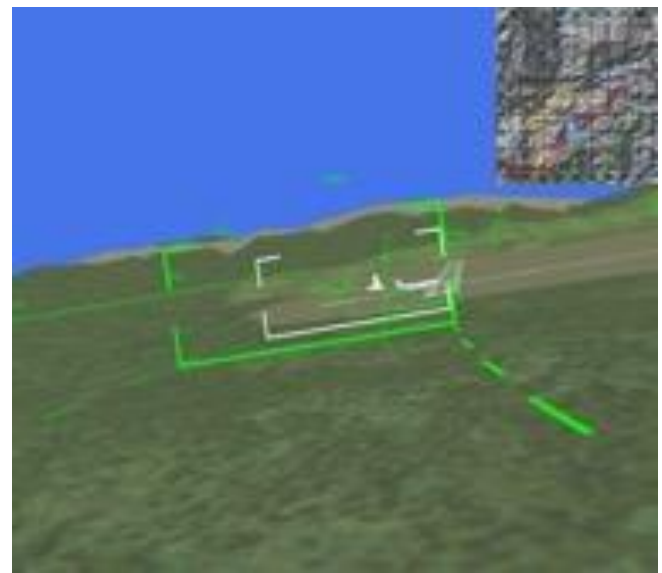
Need for graphics-based AI assistance for on-scene fire suppression supervisor: A fire predictor display system we call **FIPRED**.



Flatiron Fire (2012)

Predictor displays are valuable in several other domains.

3D Flight Path Predictor
Cone of Uncertainty
Hurricane Predictor.



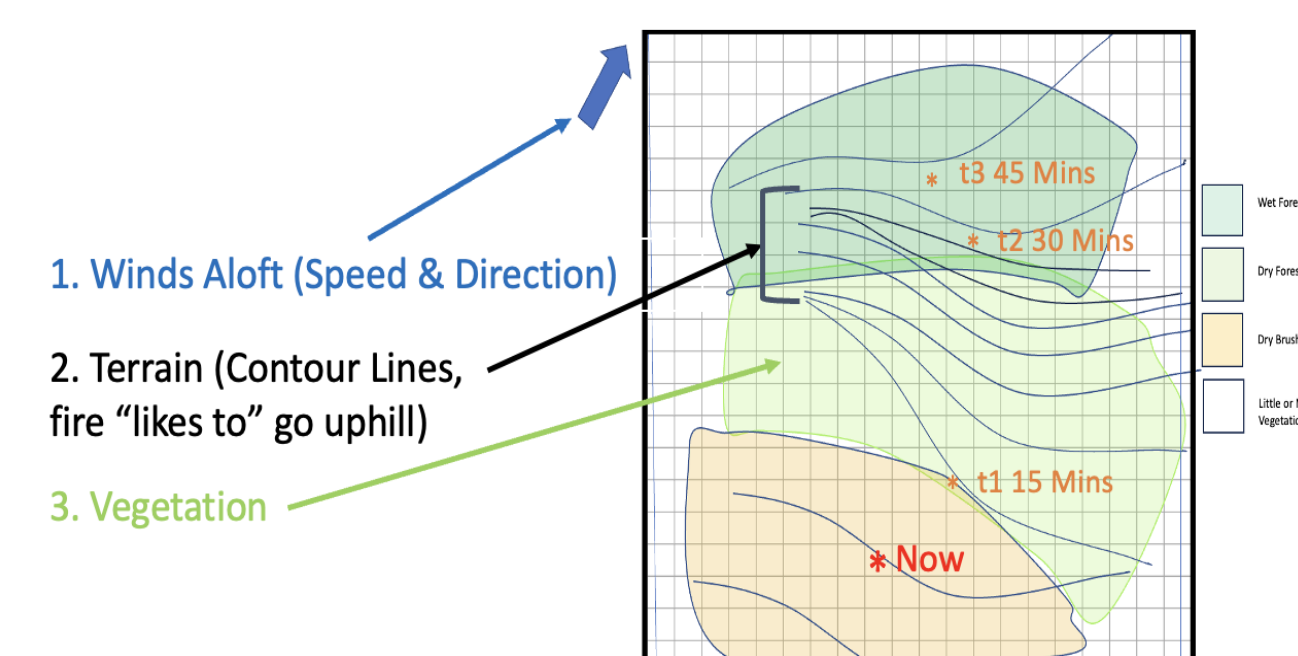
To be useful, the predictor look-ahead time (LAT) must be greater than the required action time (RAT) of responding personnel (e.g., evacuation time): **LAT > RAT**.

- Predictor displays based on an AI model of the process.
- For FIPRED, we use a simplified Fire Model (e.g., Koen, 2023).
- AI is imperfect:** what happens when the user becomes totally dependent upon the AI and it makes a mistake: **The AI fails**. The user follows its erroneous guidance.
 - This is **automation bias**.

Methods

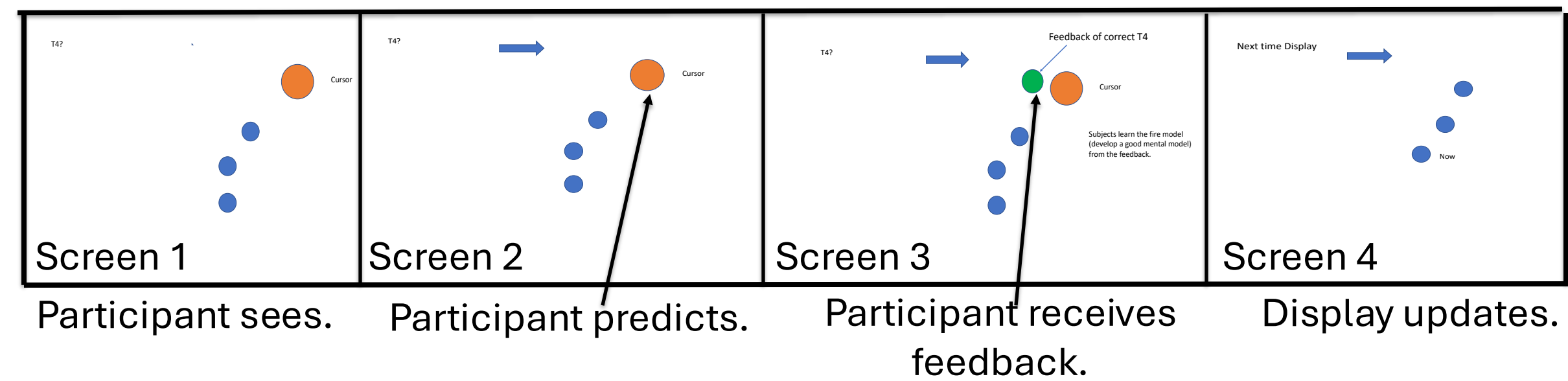
The FIPRED Display

- Simple Fire Model predicts the trajectory.
- 3 points into the future, (t1, t2, t3) based on our 3 factor Fire Model:
 - Winds aloft (speed and direction)
 - Vegetation
 - Terrain (contours): fires "like to" go uphill

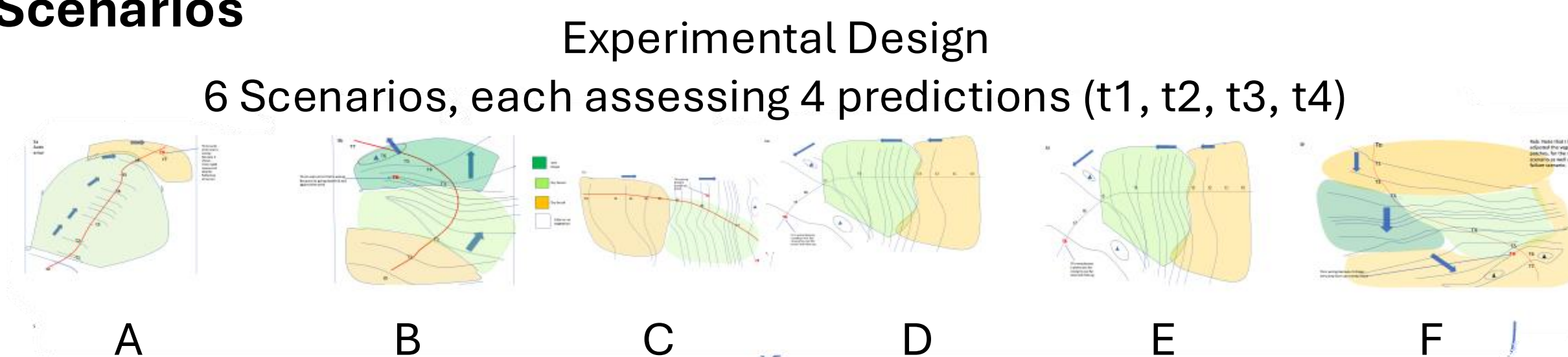


- Participants need to predict the forward point of the fire boundary at the next time point based on:
 - What the prediction says.
 - What the raw data (winds, vegetation, terrain) are telling them.

Training

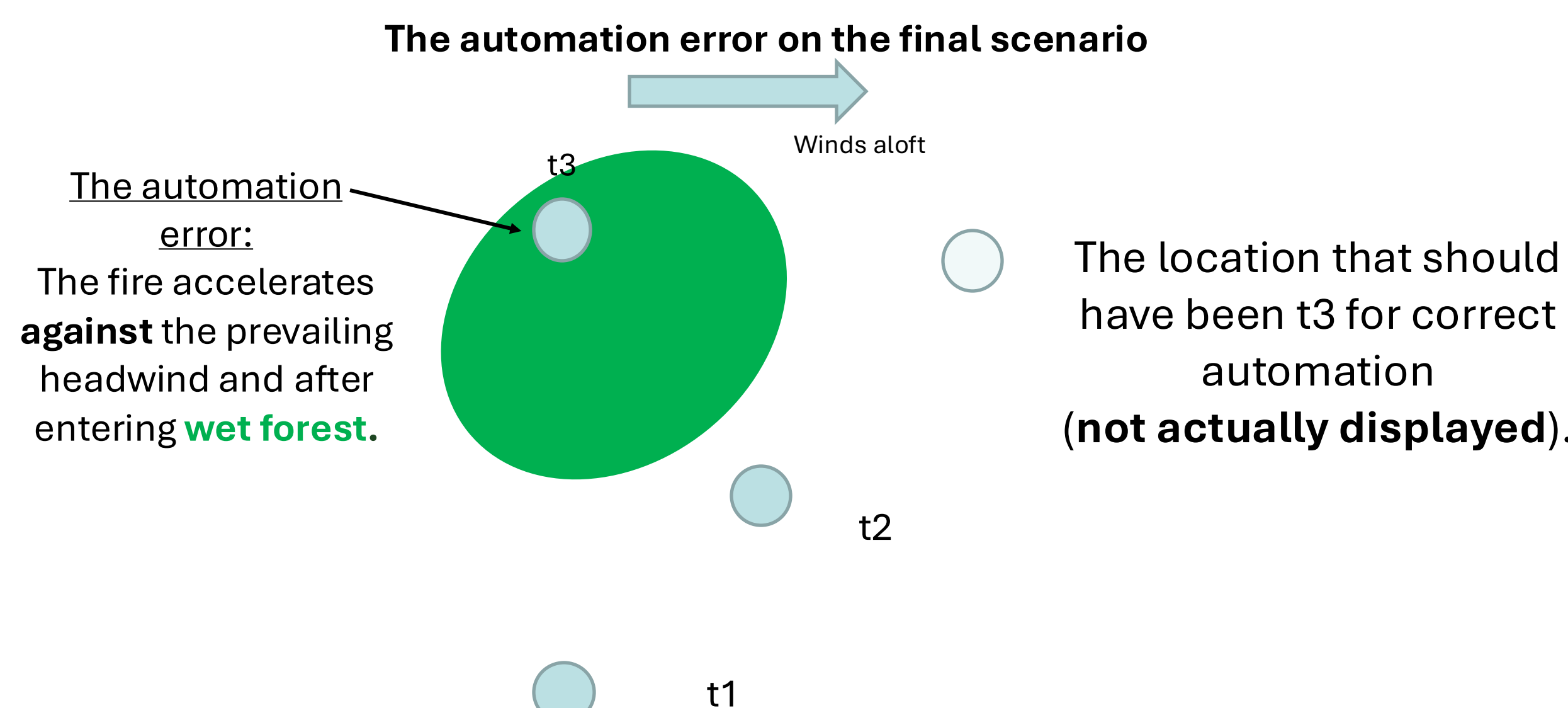


Scenarios



- 4 cycles with feedback for each fire scenario.
- Six different fire scenarios: counterbalanced order.
- Across these scenarios, participants can learn to predict (internalize the fire model).
- Automation fails on the final scenario.**

Automation Failure



Participants

22 participants: Undergraduates at CSU received course credit for 1 hour of participation. The experiment proceeded with a brief tutorial on the fire model and 4 training scenario trials.

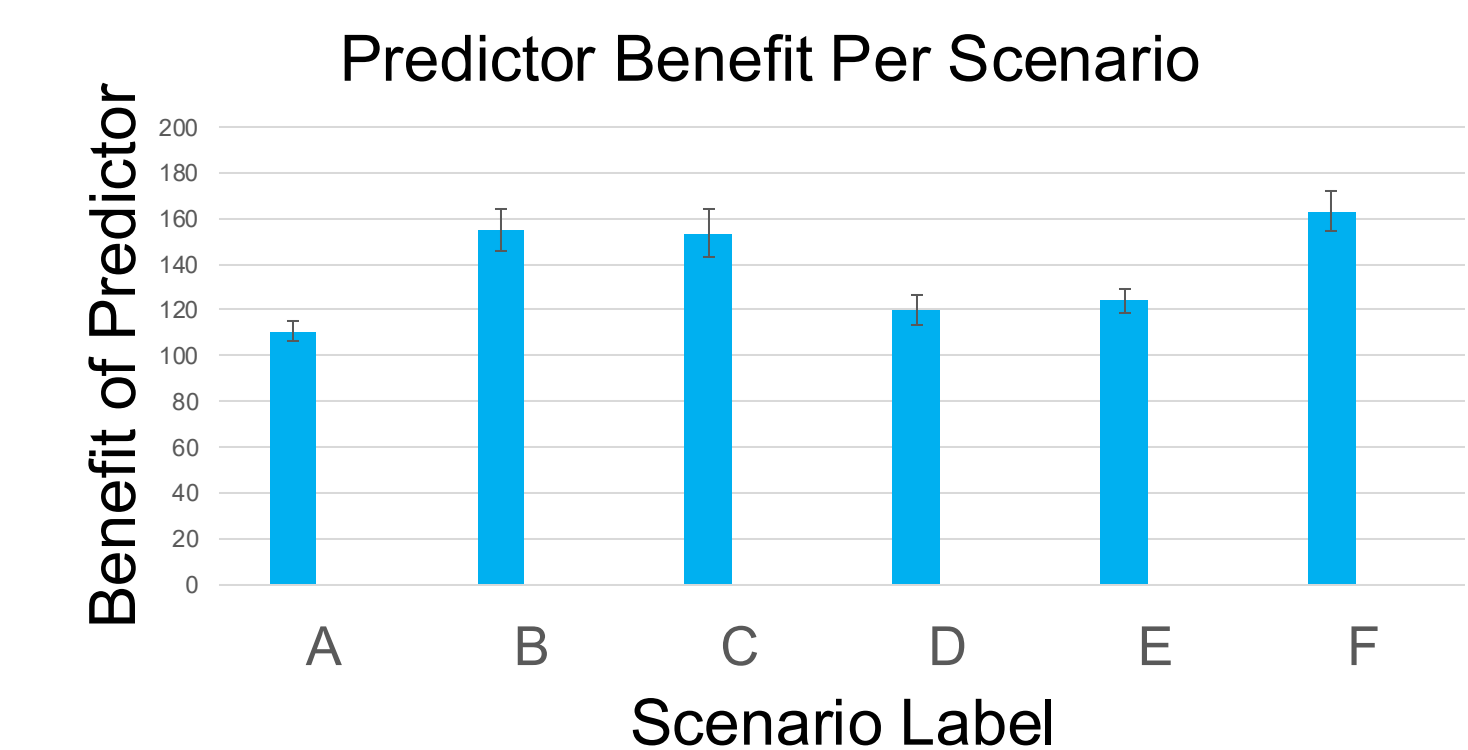
Hypotheses

- H₁:** The display will be useful to participants.
- H₂:** When the display fails, automation bias will be observed.
- H₃:** Larger dependence on automation will result in a significantly larger automation bias when automation fails.

Results

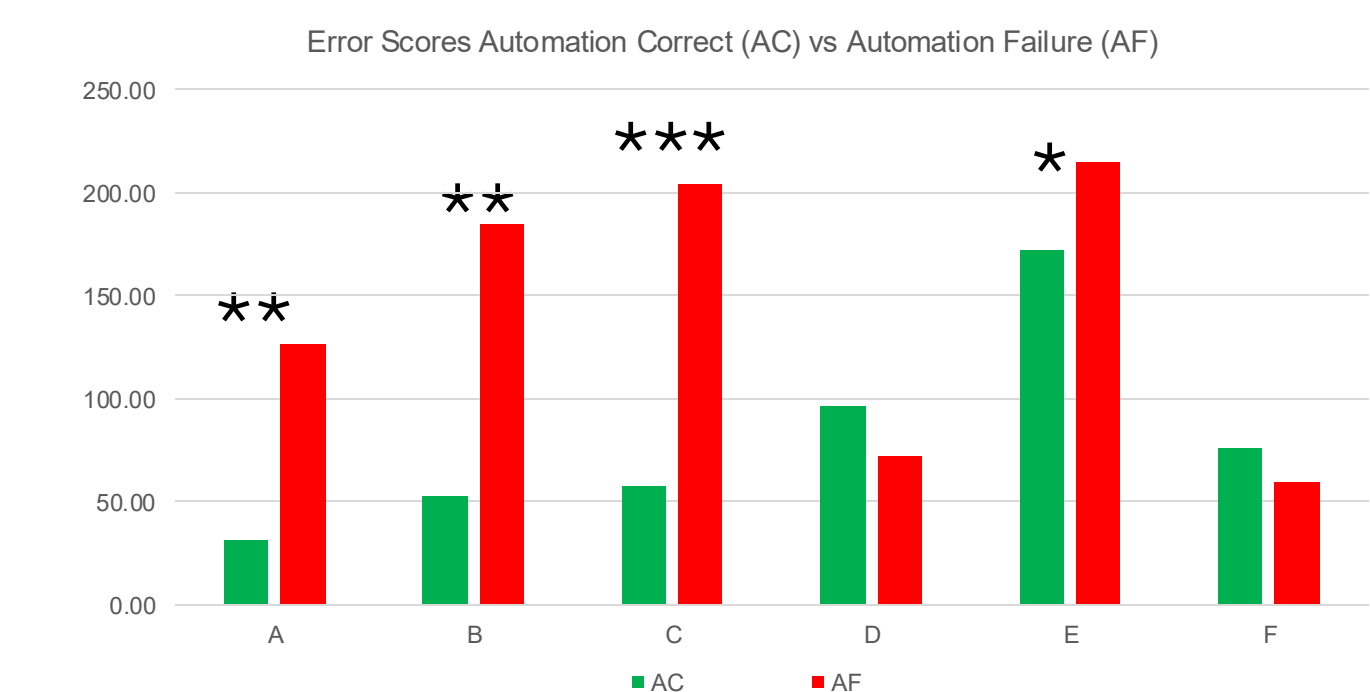
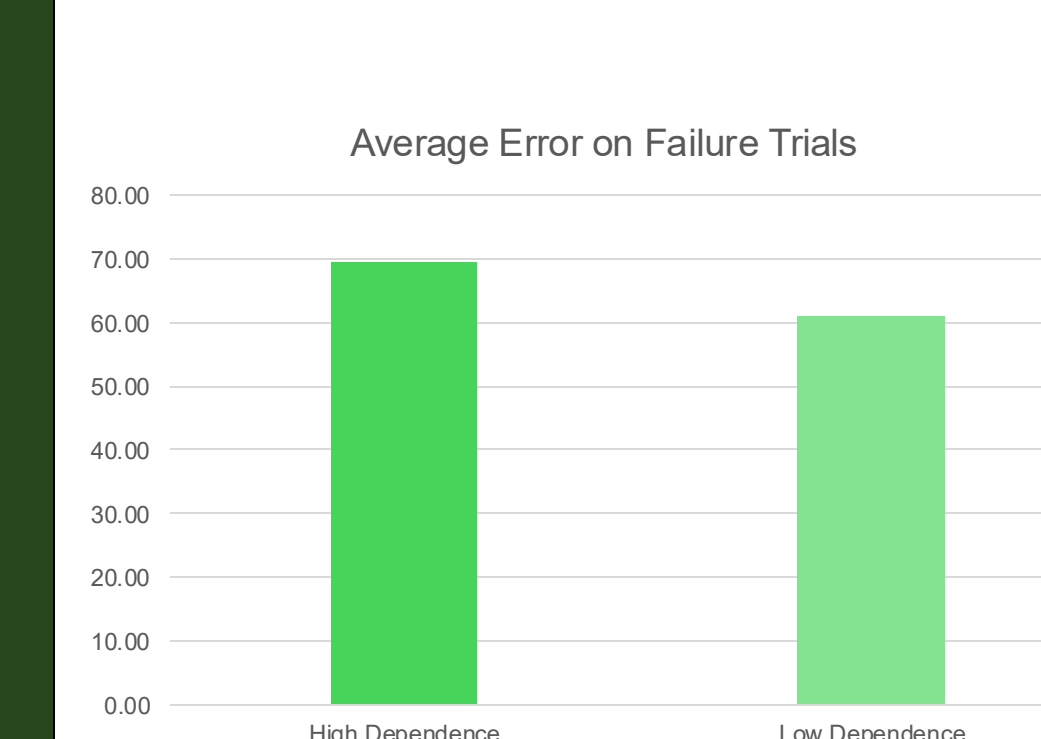
Results – H₁: Predictor Benefit – Confirmed

- We measured what the error would be if participants predicted using a simple straight-line extrapolation and compared that with their actual error. This was the benefit of the predictor display.



Results – H₂: Automation Bias – Confirmed

- Significantly larger error score when automation fails for scenarios A, B, C, & E.
 - * $p < .05$
 - ** $p < .01$
 - *** $p < .001$



Results – H₃: Dependence Does NOT Influence Automation Bias – Not Confirmed

- Participants who relied more upon the automation did NOT have a significantly larger error when it failed ($p = .24$).

Discussion & Conclusions

- We developed and evaluated a novel predictor display concept for wildfire suppression management.
 - Participants experienced a prediction performance boost from using the display.
 - This improvement led them to **automation bias**, performing worse when the automation unexpectedly failed on the final trial.
 - However, no indication of automation bias was observed between groups.
 - Certain scenarios may have had failure and correct points too close to each other.
 - Failure points may have been too obvious.
- Next steps: evaluate the benefits of presenting the raw data behind the prediction.
 - To what extent did participants depend on the raw data (terrain, winds, vegetation) to make their own predictions, *rather than the predictive display*?
 - We do not know from the current data because raw data was always displayed.
 - Next experiment: same paradigm, but raw data of terrain, winds, and vegetation are removed.

Sponsored by **Smokey Bear**.

